

Blockchain Lab #4

(Stablecoin-Specific Networks)



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Stablecoin-Specific Networks

L1 Network	Status	Key Feature
Plasma (Tether)	Mainnet beta (Sep 2025)	USDT-native, zero-fee transfers, EVM-compatible
Stable (Tether/Bitfinex)	Mainnet (Dec 2025)	USDT gas model, sub-second finality, \$28M seed round
Arc (Circle)	Testnet	USDC-native gas, built-in FX engine for USDC↔EURC
Tempo (Stripe/Paradigm)	Private testnet	Stablecoin-agnostic gas, any stablecoin as fee token
Maroo (Hashed)	In its pre-launch phase	Traditional financial institutions, domestic services
StableNet (WeMade)	In its pre-launch phase	

L2 Network	Type	Key Feature
Codex Chain	Optimism rollup	B2B enterprise settlement, USDC-native, Fireblocks integration
Giwa (Upbit)	OP Stack rollup	Global users, fintech integration (Naver Pay)
Base (Coinbase)	OP Stack rollup	Under \$0.01 fees, USDC-friendly, 142 TPS
Arbitrum	Optimistic rollup	1.5M daily transactions, deepest DeFi liquidity
Optimism	Optimistic rollup	3,044 active devs, OP Superchain ecosystem
Polygon zkEVM	ZK rollup	\$0.02 fees, enterprise-grade compliance tools
StarkNet	ZK rollup (STARK)	High throughput, AI agent payment flows
zkSync Era	ZK rollup	Native account abstraction

Plasma

Field	Value
Network Name	Plasma
Chain ID	9745
Currency Symbol	XPL
RPC Endpoint	https://rpc.plasma.to
Block Explorer	https://plasma.scan.to

Field	Value
Network Name	Plasma Testnet
Chain ID	9746
Currency Symbol	XPL
RPC Endpoint	https://testnet-rpc.plasma.to
Block Explorer	https://testnet.plasma.scan.to

Plasma

	Plasma Testnet	Sepolia
Chain ID	9746	11155111
Purpose	Deploy/test on Plasma	Test on Ethereum ecosystem
Required for Plasma dev?	Yes	No
Required for ETH bridge?	—	Yes

- Plasma is an EVM-compatible L1 built specifically for stablecoin payments. It combines zero-fee USDT transfers (gas sponsored at protocol level) with deterministic sub-second finality via PlasmaBFT consensus.
- Plasma operates as a sidechain to Bitcoin, using a custom BFT consensus mechanism inspired by HotStuff, not as an Ethereum rollup. It uses a modified Reth execution layer for EVM compatibility, but its security anchor is Bitcoin, not Ethereum.

Stable

Field	Value
Network Name	Stable
Chain ID	988
Gas Token	USDT
Governance Token	STABLE
RPC Endpoint	https://rpc.stable.xyz
Block Explorer	https://stablescan.xyz

Field	Value
Network Name	Stable Testnet
Chain ID	2201
Gas Token	USDT0
Governance Token	STABLE
RPC Endpoint (HTTP)	https://rpc.testnet.stable.xyz
RPC Endpoint (WebSocket)	wss://rpc.testnet.stable.xyz
Block Explorer	https://testnet.stablescan.xyz

Arc

Field	Value
Network Name	Arc Testnet
Chain ID	5042002
Gas Token	USDC
RPC Endpoint (HTTP)	https://rpc.testnet.arc.network
RPC Endpoint (WebSocket)	wss://rpc.drpc.testnet.arc.network
Block Explorer	https://testnet.arcscan.app

Tempo

Field	Value
Network Name	Tempo Testnet (Moderato)
Chain ID	42431
Gas Token	USD (stablecoin-based)
RPC Endpoint (HTTP)	https://rpc.moderato.tempo.xyz
RPC Endpoint (WebSocket)	wss://rpc.moderato.tempo.xyz
Block Explorer	https://explorer.tempo.xyz

Maroo

- It is positioned as a "sovereign blockchain" that combines the openness of public blockchains with built-in regulatory compliance, including AML and CFT requirements enforced at the protocol level.
- Transaction fees are paid in OKRW, a KRW-pegged stablecoin, eliminating the need for volatile native tokens and improving cost predictability.
- Maroo features a Dual-Track Transaction Model on a single Layer 1 blockchain, with an Open Path available to startups and developers for permissionless experimentation, and a Regulated Path that enforces pre-approval and compliance checks via the Programmable Compliance Layer (PCL) for transactions requiring identity verification, jurisdictional compliance, asset-type rules, or higher transaction limits.
- Maroo also plans verifiable privacy using zero-knowledge proofs as one of two candidate implementation approaches — the other being gateway-based access control — with both still subject to proof-of-concept validation.
- It additionally features a dual-path transaction processing architecture for experimental and regulated flows, and an AI-agent-dedicated account structure with KYA (Know Your Agent) verification, designed for autonomous software agents operating under compliance rules.

StableNet

Field	Value
Network Name	StableNet Testnet
Chain ID	Not yet publicly specified
Gas Token	BaseCoin (KRW-pegged stablecoind)
Block Explorer	https://explorer.stablrnet.network
Faucet	faucet.stablrnet.network

Codex Chain

Field	Value
Network Name	Codex
Chain ID	81224
Gas Token	ETH
Block Explorer	https://explorer.codex.xyz
RPC URL (HTTP)	https://rpc.codex.xyz
RPC URL (WebSocket)	wss://rpc.codex.xyz

GIWA Sepolia

Field	Value
Network Name	GIWA Sepolia
Chain ID	91342
Currency Symbol	ETH
RPC Endpoint	https://sepolia-rpc.giwa.io
Block Explorer	https://sepolia-explorer.giwa.io

Faucet	URL	Limit
Official	https://faucet.giwa.io	up to 0.005 ETH/day
Nodit	https://faucet.lambda256.io/giwa-sepolia	0.01 ETH/day

Adding to MetaMask manually

Go to Settings → Networks → Add Network → Add a Network Manually, then input the Network Name, RPC URL, Chain ID, and Currency Symbol and click Save.

Bridge

You can bridge ETH from Sepolia to GIWA Sepolia at <https://bridge-giwa.vercel.app>

What to Do Today

1. Explore StableNet.

Lecture Schedule (First 7 Weeks)

Week	Topics	Assignment
1	수업 안내, 조교 소개, 학생 소개	MetaMask 설치 & 사용법 설명
	LLM, Vibe Coding, Git	
2	Stablecoin, Micropayment, AI Agent가 만나는 지점, Web2 vs Web3 결제 구조 비교	
	MetaMask 설치, Seed 문구 개념, EOA vs Smart Contract Account	
3	트랜잭션 구조(nonce, gas, signature), Public/Private Key 이해, Testnet, Faucet 개념	
	Sepolia ETH 받기, 지갑 간 송금, Etherscan으로 트랜잭션 추적	
4	Smart Contract란? 상태(state)와 함수(function), require, msg.sender, msg.value	
	Remix로 첫 컨트랙트 배포, "Hello Blockchain"	
5	Hardhat 프로젝트 구조, 로컬 체인 vs 테스트넷, ABI 개념	
	Hardhat + MetaMask 연결, 컨트랙트 배포 스크립트 작성	
6	Token이란? balanceOf, transfer, approve, allowance, OpenZeppelin 사용 이유	
	ERC-20 토큰 직접 배포	
7	중간고사 기간	